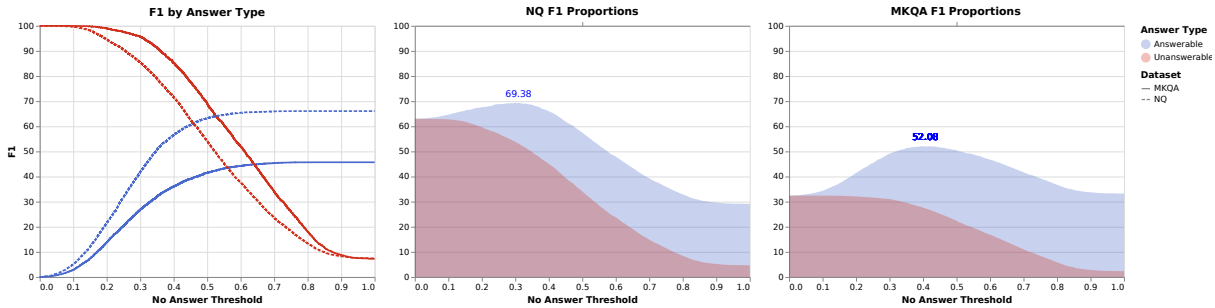

Figure 4: **Comparing MKQA and NQ English annotations.** The performance of the same English BERT-LARGE model on each of Natural Questions (NQ) annotations and MKQA annotations, using the MKQA evaluation metrics. For all plots the y-axis is F1 score and the x-axis is the value of the threshold over No Answer probabilities. F1 by Answer Type (left diagram) compares the accuracy of the model on Answerable and Unanswerable examples for each dataset, showing Unanswerable examples are on average easier in MKQA, and Answerable examples are on average harder in MKQA. NQ F1 Proportions (middle) and MKQA F1 Proportions (right) show what proportion of the aggregate F1 score is derived from each Answer Type. These plots demonstrate MKQA is more difficult than NQ because there is a higher proportion of answerable questions, which are harder on average.

English-only annotations between Natural Questions (NQ) and MKQA. In Figure 4 we use the same BERT-LARGE English model (trained on NQ, using Gold NQ passages) and evaluate it on both sets of annotations. The "F1 by Answer Type" diagram shows unanswerable examples in MKQA (red line) are easier than the unanswerable examples in NQ (red dashed line), as the model maintains higher performance at all No Answer confidence thresholds. The opposite relationship is observed for answerable examples.

We hypothesize that this is due to the *Retrieval-Independence* property and high coverage of our re-annotation process (described in Section 3). Due to the annotation procedures NQ uses, there are several cases that can lead to a potential answer missing from the dataset: (a) the initial retrieval may have not produced a candidate, (b) the answer may have not been in Wikipedia, or (c) NQ graders may have missed a valid answer. MKQA annotations are not susceptible to (a) and (b) and likely less impacted by (c). Consequently, the most challenging questions migrated from unanswerable in NQ to answerable in MKQA, shifting the unanswerable distribution from 63% to 32% (as shown in Figure 2). Consider the following examples.

(a) NQ retrieval failure In this example, the NQ retrieved document does not contain an answer to the question, causing no long or short answer (*No Answer*) in NQ. There exists a better Wikipedia document (Wheel of Fortune) that does contain the MKQA answer "Autumn Erhard".

- **Q:** *Who won the most money on wheel of fortune?*
- **NQ URL:** Wikipedia: *American game show winnings records*.
- **NQ Answer:** *No Answer*
- **MKQA Answers:** *"Autumn Erhard"*

(b) No Wikipedia answer This is also an answerable query, labelled as no answer by NQ, because the answer is not found on Wikipedia (either by NQ or our best efforts). However, an answer can be found by MKQA graders from other websites and sources.

- **Q:** *How many teeth does a saltwater crocodile have?*
- **NQ URL:** Wikipedia: *Saltwater Crocodile*.
- **NQ Answer:** *No Answer*
- **MKQA Answers:** *"66"*

(c) Annotator misses valid answer For this query, the answer is clearly visible in the provided Wikipedia article, but NQ's annotation process yields no answer.

- **Q:** *What language do they speak in the ukraine?*
- **NQ URL:** Wikipedia: *Languages of Ukraine*.
- **NQ Answer:** *No Answer*
- **MKQA Answers:** *"Ukrainian"*

Given the answer to these queries are not easily found in the corpus, by retrieval, or by human annotators, they are likely more challenging on average. As such, their label shift from no answer in NQ to answerable in MKQA likely explains why there is higher mean difficulty of answerable questions in

MKQA, as observed in Figure 4. To understand the prevalence of each error type, we compute how often *any* MKQA answer appears in the retrieved document for which the NQ label says no answer exists. We find a valid answer appears in 70.4% of these documents, suggesting category (c), annotator error, is the largest source of such unanswerable queries in NQ (and the largest source of improvement in label quality for MKQA).

The middle and right diagrams in Figure 4 normalize the answer types by their proportion within the dataset, so we can compare their relative contributions to the aggregate F1 (the sum of answerable and unanswerable). NQ labels enable a much higher aggregate F1 score (69.38% at the best threshold) than MKQA (52.08% at the best threshold) primarily due to the higher proportion of unanswerable examples — which are easier on average than answerable examples. By comparing the ratio of unanswerable to answerable examples attempted at the best thresholds in each of the middle and right diagrams (the blue regions vs. the red regions) we see that the MKQA task is more oriented to answering questions rather than abstaining.

Due to the *Parallel Question* property of MKQA, the dataset is similarly challenging in all 26 languages. There is also a noticeable gap between the performance on English and on lower-resourced languages (Figure 3). For Korean and Arabic the best F1 score is only 6% higher than the lower bound score of 32.42% obtained from predicting exclusively "unanswerable." This demonstrates that existing transfer learning methods have significant deficits to overcome for low-resource multilingual QA to match English performance. MKQA offers a challenging benchmark to measure this cross-language progress specifically.

Future Work The parallel questions property of MKQA offers alternative task setups in addition to typical open domain question answering. Lewis et al. (2020) suggests a generalized cross-lingual transfer task (G-XLT) where the question and answer languages are intentionally different. Alternatively, future work might assume we are given the English question-answer pairs, and attempt to propagate these answers into other languages by localizing the questions and answers.

We anticipate that this dataset will enable industry practitioners and researchers to rapidly test and compare novel cutting-edge techniques for QA against existing techniques in a more fair, comparable, and precise manner than previous benchmarks. Additionally, we hope that the linguistic diversity and large number of languages will inspire more researchers to treat model performance across many (partially less-resourced) languages as an important and worthy goal in itself. As MKQA offers the **only** open-QA option for many of these languages, we also hope to spark important research in these monolingual, non-English settings.

7 Conclusion

In this work, we introduce a multilingual open domain question answering evaluation set. Its properties, including geographical invariance, language-parallel questions, retrieval-independent annotations, and linguistic diversity, set it apart from existing resources in terms of annotation quality, difficulty, and flexibility to evaluate new approaches. We encourage future multilingual benchmarks to adopt data collection and annotation principles to promote higher-quality, and informative evaluation practices. We evaluate several baselines, based on state-of-the-art methods, and demonstrate ample room for improvement both in English and in the tail of lower-resourced languages. We hope that this evaluation set enables wider exploration of cross-lingual and monolingual methods in non-English QA.

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References

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. 2020a. Translation artifacts in cross-lingual transfer learning. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 7674–7684.

Mikel Artetxe, Sebastian Ruder, and Dani Yogatama. 2020b. On the cross-lingual transferability of monolingual representations. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4623–4637.

Akari Asai, Jungo Kasai, Jonathan H Clark, Kenton Lee, Eunsol Choi, and Hannaneh Hajishirzi. 2021. Xor qa: Cross-lingual open-retrieval question answering. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 547–564.

Danqi Chen, Jason Bolton, and Christopher D. Manning. 2016. A thorough examination of the CNN/daily mail reading comprehension task. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2358–2367, Berlin, Germany. Association for Computational Linguistics.

Danqi Chen, Adam Fisch, Jason Weston, and Antoine Bordes. 2017. Reading Wikipedia to answer open-domain questions. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1870–1879, Vancouver, Canada. Association for Computational Linguistics.

Jonathan H Clark, Eunsol Choi, Michael Collins, Dan Garrette, Tom Kwiatkowski, Vitaly Nikolaev, and Jennimaria Palomaki. 2020. Tydi qa: A benchmark for information-seeking question answering in typologically diverse languages. *Transactions of the Association for Computational Linguistics*, 8:454–470.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Édouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451.

Yiming Cui, Wanxiang Che, Ting Liu, Bing Qin, Shijin Wang, and Guoping Hu. 2019a. Cross-lingual machine reading comprehension. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 1586–1595.

Yiming Cui, Zhipeng Chen, Si Wei, Shijin Wang, Ting Liu, and Guoping Hu. 2017. Attention-over-attention neural networks for reading comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 593–602, Vancouver, Canada. Association for Computational Linguistics.

Yiming Cui, Ting Liu, Wanxiang Che, Li Xiao, Zhipeng Chen, Wentao Ma, Shijin Wang, and Guoping Hu. 2019b. A span-extraction dataset for chinese machine reading comprehension. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 5886–5891.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186.

Bhuwan Dhingra, Hanxiao Liu, Zhilin Yang, William Cohen, and Ruslan Salakhutdinov. 2017. Gated-attention readers for text comprehension. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1832–1846, Vancouver, Canada. Association for Computational Linguistics.

B Green, A Wolf, C Chomsky, and K Laughery. 1986. *BASEBALL: An Automatic Question Answerer*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA.